

# SIDS Cancer Registry Comprehensive Toolkit

Implementation Guide · Data Preparation · Statistical Methods · Policy Reporting

**Version:** 1.0 — 2026  
**Reference Registry:** Barbados National Cancer Registry (BNR)  
**Coverage:** Incidence 2013–2022 · Mortality 2008–2024  
**Standard:** WHO 2000 Age-standardisation · IARC ICD-10  
**Audience:** Registry Staff · Epidemiologists · Policymakers

OVERVIEW

INCIDENCE

MORTALITY

SURVIVAL

PREVALENCE

PROJECTIONS

DATA QUALITY

REPORTS

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## PART I — FRAMEWORK OVERVIEW & CONTEXT

### 1.1 About This Toolkit

A comprehensive reference for setting up, running and reporting from a SIDS cancer registry dashboard

This toolkit accompanies the **SIDS Cancer Registry Dashboard**, an open-source R/Shiny application developed from the Barbados National Cancer Registry (BNR). It provides everything needed to install, configure, populate and report from a population-based cancer surveillance system adapted for any Small Island Developing State.

|                                                                 |                                                                |                                                                |                                                         |
|-----------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| <b>DASHBOARD MODULES</b><br><b>8</b><br><br>analytical sections | <b>DATA FILES REQUIRED</b><br><b>5</b><br><br>CSV / XLSX / SHP | <b>STATISTICAL METHODS</b><br><b>6</b><br><br>WHO-standardised | <b>REPORT SLIDES</b><br><b>15</b><br><br>auto-generated |
|-----------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|

### Who Should Use This Toolkit

| Registry Staff & Data Managers  | Sections 2, 4 and 6 — data preparation, file setup and configuration.            |
|---------------------------------|----------------------------------------------------------------------------------|
| Epidemiologists & Analysts      | Sections 3 and 6 — statistical methods, interpretation and adaptation.           |
| Ministry of Health Policymakers | Section 5 — reading dashboard outputs, key indicators and report interpretation. |

### 1.2 SIDS-Specific Considerations

Cancer registries in Small Island Developing States face a unique set of challenges that this framework is designed to address. Understanding these constraints is essential before configuring the dashboard.

#### ■ Small Population Denominators

Rates based on fewer than 20 cases are statistically unstable. The dashboard flags such strata and the projection module automatically applies a linear fallback when the negative binomial model fails to converge.

#### ■ Out-migration & Follow-up Loss

Island populations experience high emigration rates, leading to incomplete survival follow-up. The prevalence module applies a conservative 12-month last-contact window ( $dlc \geq prevalence\_date - 365$ ) to classify survivors.

#### ■ Dual Disease Burden

Many SIDS registries record both communicable and non-communicable disease impacts. Prostate, breast, cervical and colorectal cancers dominate SIDS incidence; the dashboard's top-site logic dynamically identifies these from your data.

#### ■ Limited Registry Infrastructure

The framework uses only open-source R libraries and flat CSV/XLSX files — no database server required. It can run on a standard laptop or on shinyapps.io free tier.

■ **Age Structure Differences**

SIDS often have younger or more rapidly ageing populations than high-income countries. The WHO 2000 standard population is used throughout for comparability with GLOBOCAN and other international benchmarks.

**1.3 Dashboard Architecture**

The dashboard is structured as a single-file R Shiny application with three logical layers:

| Data Layer         | CSV incidence & mortality records; WPP population XLSX; parish shapefiles | readxl, sf, dplyr, tidyr, purrr                          |
|--------------------|---------------------------------------------------------------------------|----------------------------------------------------------|
| Analytical Layer   | ASIR, ASMR, cumulative incidence, KM survival, prevalence, projections    | survival, MASS, lubridate, qcc                           |
| Presentation Layer | Sidebar navigation, interactive charts, tables, maps, PPT export          | shiny, shinydashboard, plotly, leaflet, DT, officer, rvg |

■■ Authentication is handled by **shinyauthr** with bcrypt password hashing via the **sodium** package. Credentials are stored in a local tibble — replace with a database-backed user table for production deployments with multiple users.

## PART II — DATA PREPARATION GUIDE

# Data Preparation Guide

How to structure, code and clean your registry data files

## 2.1 Required Data Files

Five files are required. All paths are relative to the project root.

| File               | Path                              | Format | Description                          |
|--------------------|-----------------------------------|--------|--------------------------------------|
| Incidence data     | data/cancer_YYYY_YYYY_[reg].csv   | CSV    | One row per incident cancer case     |
| Mortality data     | data/cancer_death_YYYY_YYYY.csv   | CSV    | One row per cancer death             |
| Population (WPP)   | data/WPP.xlsx                     | XLSX   | UN WPP estimates; one sheet per year |
| Parish population  | data/parish_population_by_age.csv | CSV    | Sub-national 10-yr age bands         |
| Boundary shapefile | data/[Country]_Admin.shp          | SHP    | Administrative polygons (EPSG:4326)  |

## 2.2 Incidence Data: Required Variables

| Variable      | Type    | Description / Coding                                                 |
|---------------|---------|----------------------------------------------------------------------|
| age           | Integer | Age at diagnosis in completed years (0–100+)                         |
| sex           | String  | "male" or "female" (lowercase)                                       |
| dxyr          | Integer | Year of diagnosis (e.g. 2018)                                        |
| siteiarc      | String  | IARC cancer site label (e.g. "Prostate", "Breast (female)")          |
| top           | String  | ICD-10 topography code (e.g. "C61")                                  |
| primarysite   | String  | ICD-10 topography code (may duplicate top)                           |
| basis         | String  | Basis of diagnosis: "Histology", "Cytology", "DCO", "Clinical", etc. |
| IncidenceDate | String  | Date of diagnosis: YYYYMMDD or "DD MMM YYYY" format                  |
| deceased      | String  | "dead" or "alive"                                                    |
| dod           | String  | Date of death (DD-MM-YYYY); empty if alive                           |
| dlc           | String  | Date of last contact (DD-MM-YYYY)                                    |
| parish        | String  | Sub-national administrative unit (standardised name)                 |

■■ Parish names must be consistent across the incidence file, mortality file, population CSV and shapefile NAME\_1 field. Use the **standardize\_parish\_names()** function to harmonise abbreviations (e.g. "St. Michael" → "Saint Michael").

## 2.3 Mortality Data: Required Variables

| Variable | Type    | Description                                                  |
|----------|---------|--------------------------------------------------------------|
| age      | Integer | Age at death in completed years                              |
| sex      | String  | "Male" or "Female" (capital first letter for mortality data) |
| dodyear  | Integer | Year of death (e.g. 2019)                                    |
| siteiarc | String  | Underlying cause of death — IARC site label                  |

|        |        |                                               |
|--------|--------|-----------------------------------------------|
| parish | String | Sub-national administrative unit of residence |
|--------|--------|-----------------------------------------------|

■ **Sex coding differs** between incidence and mortality files in the BNR reference implementation: incidence uses lowercase ("male"/"female"); mortality uses title case ("Male"/"Female"). Ensure your `compute_asmr()` filter matches your mortality file.

## 2.4 Population Data (UN World Population Prospects)

Download the WPP single-year estimates for your country from [population.un.org](https://population.un.org). Each Excel sheet represents one calendar year and must contain the following columns:

| Column  | Description                                             |
|---------|---------------------------------------------------------|
| age5    | Age group index 1–18 (1 = 0–4 yrs, 18 = 85+ yrs)        |
| sex     | "male" or "female" (lowercase)                          |
| pop_wpp | Mid-year population estimate for that age-sex-year cell |

The dashboard iterates over sheets using `map2_dfr(seq_along(years), years, ~read_excel(...))`. Sheet order must match the `years` vector defined at the top of `app.R`.

## 2.5 Geographic Boundary Files

Obtain administrative boundary shapefiles from **GADM** ([gadm.org](https://gadm.org)) or your national mapping agency. The shapefile must:

- Be projected in EPSG:4326 (WGS84 geographic coordinates) — the dashboard calls `st_transform(shapefile, 4326)` automatically.
- Contain a field named **NAME\_1** with the administrative unit names matching those in your registry data (after name standardisation).
- Pass `st_make_valid()` — run this after loading to fix geometry errors.

```
# Load and validate shapefile
parish_shapefile <- st_read("data/[Country]_Admin.shp")
st_crs(parish_shapefile) <- 3857
parish_shapefile <- st_transform(parish_shapefile, 4326)
parish_shapefile <- st_make_valid(parish_shapefile)

# Verify join key matches registry data
setdiff(unique(data$parish), parish_shapefile$NAME_1)
```

## 2.6 Data Cleaning Checklist

Complete this checklist before loading data into the dashboard:

### Incidence

- ■ No duplicate case IDs (one row per patient per primary tumour)
- ■ Age values in range 0–110 (no negatives, no 999)
- ■ Sex coded as "male" / "female" (lowercase, no missing)
- ■ dxyr within the stated incidence period (e.g. 2013–2022)
- ■ siteiarc contains recognised IARC labels; "Other and unspecified (O&U;)" used for unclassifiable
- ■ Basis coded from accepted vocabulary (Histology, Cytology, Lab, Haem, DCO, Clinical, etc.)
- ■ IncidenceDate parseable as YYYYMMDD or DD MMM YYYY
- ■ Parish names harmonised using `standardize_parish_names()`
- ■ Deceased / dod / dlc logically consistent (dead → dod not missing)

## Mortality

- ■ One row per death record; no duplicate entries
- ■ Sex coded as "Male" / "Female" (title case)
- ■ dodyear within the stated mortality period (e.g. 2008–2024)
- ■ siteiarc populated (not blank); O&U; used where site unknown
- ■ Age at death plausible (0–110)

## Population (WPP)

- ■ Correct number of sheets (one per year in your range)
- ■ age5 index runs 1–18 without gaps
- ■ pop\_wpp values > 0 for all cells; no NAs
- ■ Sheet order matches the years vector in app.R

## Geography

- ■ NAME\_1 in shapefile matches all parish values in incidence & mortality data
- ■ st\_make\_valid() returns no warnings
- ■ CRS confirmed as EPSG:4326 after transform

## PART III — STATISTICAL METHODS REFERENCE

# Statistical Methods Reference

Formulae, assumptions and R implementation for all six analytical methods

## 3.1 Age-Standardised Incidence Rate (ASIR)

The ASIR adjusts for differences in age structure between populations, enabling valid comparisons across years and between islands. The WHO 2000 World Standard Population is used throughout.

| Formula                                       | Components                                                                             |
|-----------------------------------------------|----------------------------------------------------------------------------------------|
| $ASIR = \sum (r_i \times w_i) \times 100,000$ | $r_i$ = age-specific rate in group $i$ ; $w_i$ = WHO 2000 weight for group $i$         |
| $r_i = C_i / P_i$                             | $C_i$ = observed cases in age group $i$ ; $P_i$ = mid-year population in age group $i$ |

### WHO 2000 Standard Population Weights (18 Age Groups)

| Age Group | Index | WHO 2000 Weight | Age Group | Index | WHO 2000 Weight |
|-----------|-------|-----------------|-----------|-------|-----------------|
| 0–4       | 1     | 0.08860         | 45–49     | 10    | 0.06040         |
| 5–9       | 2     | 0.08690         | 50–54     | 11    | 0.05380         |
| 10–14     | 3     | 0.08590         | 55–59     | 12    | 0.04550         |
| 15–19     | 4     | 0.08470         | 60–64     | 13    | 0.03720         |
| 20–24     | 5     | 0.08220         | 65–69     | 14    | 0.02960         |
| 25–29     | 6     | 0.07930         | 70–74     | 15    | 0.02210         |
| 30–34     | 7     | 0.07610         | 75–79     | 16    | 0.01520         |
| 35–39     | 8     | 0.07150         | 80–84     | 17    | 0.00900         |
| 40–44     | 9     | 0.06590         | 85+       | 18    | 0.00600         |

■ ■ Weights sum to 1.00 (100,000 / 100,000). The BNR implementation stores these as a named vector: **who\_weights**  
 <- c(8860,8690,...,600) / 100000

### R Implementation: compute asir()

```
compute_asir <- function(cancer_data, pop_data, who_weights, site, sex_group) {
  # 1. Filter by site and sex
  # 2. Assign 5-yr age group index (cut() with 18 bands)
  # 3. Count cases per year × age group; expand to complete grid
  # 4. Merge with WPP population; compute age-specific rate
  # 5. ASIR = sum(age_rate * who_weights[age_group]) per year
  full_df %>% group_by(year) %>%
    summarise(asir = sum(age_rate * who_weights[age_group]))
}
```

## 3.2 Age-Standardised Mortality Rate (ASMR)

The ASMR uses an identical formula to the ASIR but replaces incident cases with death counts and uses the **dodyear** field from the mortality file. The function **compute\_asmr()** mirrors **compute\_asir()** with these substitutions:

- cancer\_data → mortality\_data
- dxyr → dodyear



- Sex filter: mortality file uses title-case "Male"/"Female" vs incidence "male"/"female"

■ ■ For small island registries with fewer than 10 deaths per age group per year, age-specific rates become highly unstable. Consider aggregating to 3-year or 5-year rolling periods before computing ASMR.

### 3.3 Cumulative Incidence (0–74 years)

Cumulative incidence estimates the probability (%) that an individual develops cancer between birth and age 74, assuming no competing causes of death.

| Formula                                                      | Description                                                                            |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------|
| $CI_{0-74} = [\sum (r_{age} \times 5) / 100,000] \times 100$ | Sum over age groups 1–15 (0–4 to 70–74 yrs); multiply each rate by 5-yr interval width |

■ ■ Cumulative incidence is expressed as a percentage (%). A value of 12.5% means 12.5 in every 100 people would be expected to develop cancer before age 75.

### 3.4 Kaplan-Meier Survival Analysis

The survival module uses the **survival** package to estimate overall and sex-stratified survival probabilities from time of diagnosis.

| Parameter          | Definition in BNR Dashboard                                    |
|--------------------|----------------------------------------------------------------|
| Time origin        | Date of diagnosis (IncidenceDate, parsed by parse_incidence()) |
| Event              | Death (deceased == "dead")                                     |
| Censoring          | Alive at last contact (dlc); or lost to follow-up              |
| Stratification     | By sex; optionally by age band or cancer site                  |
| 1/3/5-yr estimates | Extracted from survfit object at t = 12, 36, 60 months         |
| Log-rank test      | survdiff() comparing male vs female survival curves            |

```
km_fit <- survfit(Surv(time_months, event) ~ sex, data = cancer_df)
summary(km_fit, times = c(12, 36, 60)) # 1, 3, 5-year survival
survdiff(Surv(time_months, event) ~ sex, data = cancer_df) # log-rank
```

### 3.5 Point Prevalence Estimation

Prevalence counts individuals alive with a cancer diagnosis on a specified reference date (default: 31 December of the last incidence year).

| Criterion                             | Classification                   | Code                       |
|---------------------------------------|----------------------------------|----------------------------|
| death_date <= prevalence_date         | Dead — exclude                   | alive_at_prev_date = FALSE |
| last_contact >= prevalence_date       | Alive — include                  | alive_at_prev_date = TRUE  |
| last_contact >= prevalence_date - 365 | Likely alive — include           | alive_at_prev_date = TRUE  |
| Otherwise                             | Unknown — exclude (conservative) | alive_at_prev_date = FALSE |

Prevalence Rate (%) = (Survivors / Total population) × 100 Population is sourced from UN WPP for the prevalence year.

### 3.6 Projection Models (Negative Binomial)

Future cancer incidence is projected using a negative binomial regression of annual case counts on calendar year, with automatic fallback to a linear model.

| Step | Action                                                       | Function                                                                |
|------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| 1    | Summarise annual observed counts per site                    | <code>group_by(dxyr) %&gt;% summarise(cases=n())</code>                 |
| 2    | Fit NB model: $\text{cases} \sim \text{year}$                | <code>MASS::glm.nb(cases ~ year, control=glm.control(maxit=200))</code> |
| 3    | Predict on observed + future years                           | <code>predict(mod, newdata=all_yrs, type="link", se.fit=TRUE)</code>    |
| 4    | Back-transform: $\exp(\text{fit} \pm 1.96 \times \text{se})$ | 95% prediction interval on count scale                                  |
| 5    | Fallback if NB fails to converge                             | <code>lm(cases ~ year)</code> with prediction interval                  |

■ ■ For sites with fewer than 30 observed cases in total, projection intervals will be very wide and should not be used for planning. Report only totals and top-3 sites for registries with <200 annual cases.

PART IV — IMPLEMENTATION GUIDE

# Implementation Guide

Installing, configuring and deploying the dashboard on a new island registry

## 4.1 Software Prerequisites

| Software              | Minimum Version | Notes                                          |
|-----------------------|-----------------|------------------------------------------------|
| R                     | ≥ 4.2.0         | Download from cran.r-project.org               |
| RStudio (recommended) | ≥ 2023.06       | Or any R IDE; VS Code + R extension also works |
| shiny                 | ≥ 1.7           | install.packages("shiny")                      |
| shinydashboard        | ≥ 0.7           | install.packages("shinydashboard")             |
| shinyauthr            | ≥ 1.0           | install.packages("shinyauthr")                 |
| sodium                | ≥ 1.3           | install.packages("sodium")                     |
| survival              | ≥ 3.5           | Usually pre-installed with R                   |
| MASS                  | ≥ 7.3           | Usually pre-installed with R                   |
| officer + rvg         | ≥ 0.6 / 0.6     | Required for PowerPoint export                 |
| sf + leaflet          | ≥ 1.0 / 2.1     | Required for choropleth maps                   |
| plotly, DT, viridis   | Any recent      | install.packages(c("plotly","DT","viridis"))   |

```
# Install all dependencies in one block
libs <- c("shiny","shinydashboard","shinyauthr","shinyjs","sodium",
          "ggplot2","DT","lubridate","survival","plotly","tidyr","purrr",
          "readxl","qcc","officer","rvg","sf","leaflet","viridis","dplyr")
installed_libs <- libs %in% rownames(installed.packages())
if (any(!installed_libs)) install.packages(libs[!installed_libs])
```

## 4.2 File & Directory Structure

```
registry-dashboard/
■■■■ app.R                # Main Shiny application file
■■■■ www/
■   ■■■■ logo.png         # Registry logo (displayed in header)
■■■■ data/
    ■■■■ cancer_2013_2022_reg.csv    # Incidence data
    ■■■■ cancer_death_2008_2024.csv  # Mortality data
    ■■■■ WPP.xlsx                  # UN WPP population estimates
    ■■■■ parish_population_by_age.csv
    ■■■■ [Country]_Admin.shp        # Shapefile
    ■■■■ [Country]_Admin.dbf
    ■■■■ [Country]_Admin.prj
    ■■■■ [Country]_Admin.shx
```

## 4.3 Configuration Parameters

At the top of app.R, update these parameters for your registry:

| Parameter            | Location in app.R        | Example Value |
|----------------------|--------------------------|---------------|
| Incidence year range | years vector / UI labels | 2015:2023     |
| Mortality year range | mortality years vector   | 2008:2024     |

|                       |                            |                                  |
|-----------------------|----------------------------|----------------------------------|
| Shapefile path        | st_read() call             | "data/Jamaica_Parish.shp"        |
| Shapefile join column | create_parish_map()        | "NAME_1" or "DIST_NAME"          |
| Admin unit name       | standardize_parish_names() | Replace "parish" with "district" |
| Dashboard title       | dashboardHeader(title=...) | "Jamaica Cancer Registry"        |
| Registry logo         | tags\$link(rel="icon",...) | "www/jcr_logo.png"               |
| Data file paths       | read.csv() calls           | "data/cancer_2015_2023_jcr.csv"  |

## 4.4 Authentication Setup

The dashboard uses **shinyauthr** with bcrypt-hashed passwords. To create or update credentials:

```
# Generate a hashed password (run once in R console)
library(sodium)
sodium::password_store("your_secure_password_here")
# Copy the output hash into user_base below

# In app.R – update user_base:
user_base <- tibble::tibble(
  user      = c("analyst1", "manager1"),
  password = c(sodium::password_store("pass1"),
               sodium::password_store("pass2"))
)
```

■ Never store plain-text passwords. The `sodium::password_store()` function applies bcrypt hashing with a random salt — each call produces a different hash string, but `sodium::password_verify()` will still validate correctly.

## 4.5 Deploying to shinyapps.io

- **Step 1:** Create a free account at **shinyapps.io** (Posit).
- **Step 2:** Install `rsconnect`: `install.packages("rsconnect")`
- **Step 3:** Set your credentials: `rsconnect::setAccountInfo(name, token, secret)`
- **Step 4:** Deploy: `rsconnect::deployApp(appDir = "registry-dashboard/", appName = "cancer-dashboard")`
- **Step 5:** Set app visibility to **Private** in the shinyapps.io dashboard to restrict access.
- **Step 6:** Free tier allows 25 active hours/month; upgrade to Standard for production use.

■ Do NOT store the `data/` folder contents in a public GitHub repository. Add `data/` to your `.gitignore`. Upload data files separately or use environment variables and a secure storage bucket (e.g. AWS S3) for production.

## PART V — POLICY REPORTING GUIDE

# Policy Reporting Guide

Reading dashboard outputs, key indicators and generating the PowerPoint report

## 5.1 Key Indicators for Policymakers

The dashboard produces six categories of indicators. This table explains what each measures and how to interpret the direction of change.

| Indicator                 | Module       | Unit            | Higher is...          | Benchmark Context                                 |
|---------------------------|--------------|-----------------|-----------------------|---------------------------------------------------|
| ASIR                      | Incidence    | /100,000        | More disease burden   | Compare to GLOBOCAN regional average              |
| ASMR                      | Mortality    | /100,000        | More deaths           | Higher ratio ASMR:ASIR = poorer prognosis         |
| Cumulative Incidence 0–74 | Incidence    | %               | Greater lifetime risk | Global avg ~20% all cancers combined              |
| 5-Year Survival           | Survival     | %               | Better outcomes       | High-income countries: 60–80% (site-specific)     |
| Prevalence Rate           | Prevalence   | % of population | More survivors        | Increases with better survival & rising incidence |
| MV%                       | Data Quality | %               | Better data quality   | IARC target: >75%                                 |
| DCO%                      | Data Quality | %               | Worse data quality    | IARC target: <5%                                  |

## 5.2 Interpreting Dashboard Outputs

### ■ Rising ASIR with stable ASMR

Suggests improving survival — more cases being diagnosed and surviving. May reflect better screening uptake, earlier diagnosis or improved treatment access.

### ■ Rising ASIR with rising ASMR

Indicates a genuine increase in cancer burden without survival improvement. May signal gaps in treatment infrastructure, late-stage presentation or risk factor exposure.

### ■ Low MV% (<60%)

Indicates many cases lack histological confirmation — possibly because patients are diagnosed clinically or at death. Affects ASIR reliability and site-specific comparisons.

### ■ High DCO% (>10%)

Many cases are registered only from death certificates, suggesting the registry is missing live diagnoses. Active case-finding and hospital linkage improvements are needed.

### ■ Wide projection intervals

Expected for small island registries with few cases per site. Report only totals and top-3 sites; avoid site-specific projections for rare cancers (<20 cases/year).

## 5.3 PowerPoint Report Generation

The Reports module generates a 15-slide PowerPoint file using the **officer** and **rvg** packages. All charts are embedded as editable vector graphics.

| Slide | Title                    | Content                                            |
|-------|--------------------------|----------------------------------------------------|
| 1     | Title Slide              | Registry name and report date                      |
| 2     | Data Quality Indicators  | MV%, DCO%, Ill-defined sites%                      |
| 3     | Key Statistics           | Total cases, deaths, mean age at diagnosis & death |
| 4     | Age Distribution         | Bar chart: Paediatric / Adult / Elderly splits     |
| 5     | Trend in Incident Cases  | Annual bar chart (incidence period)                |
| 6     | Top 10 Cancer Sites      | Horizontal bar chart by count                      |
| 7     | ASIR by Sex              | Line chart: Male vs Female trend                   |
| 8     | ASIR — Top 5 Sites       | Multi-line ASIR trends per site                    |
| 9     | Cancer Deaths Trend      | Annual bar chart (mortality period)                |
| 10    | ASMR by Sex              | Line chart: Male vs Female ASMR                    |
| 11    | Top 10 Mortality Sites   | Horizontal bar chart by deaths                     |
| 12    | ASMR — Top 5 Fatal Sites | Multi-line ASMR trends per site                    |
| 13    | Survival Curves          | KM chart by sex; log-rank p-value                  |
| 14    | Prevalence Summary       | Survivor count, rate, age-sex bar chart            |
| 15    | Projections              | All cancers observed + projected with 95% PI       |

## 5.4 Data Quality Benchmarks (IARC)

| Indicator           | Excellent | Acceptable | Poor | Action if Poor                                         |
|---------------------|-----------|------------|------|--------------------------------------------------------|
| MV%                 | >85%      | 75–85%     | <75% | Improve pathology linkage; train staff in basis coding |
| DCO%                | <2%       | 2–5%       | >5%  | Active case-finding; hospital records linkage          |
| Ill-defined sites   | <5%       | 5–10%      | >10% | ICD-10 coding training; medical record review          |
| Proportion with age | >98%      | 95–98%     | <95% | Mandatory age field in admission forms                 |

## PART VI — ADAPTATION FOR OTHER SIDS

# Adaptation for Other SIDS

Step-by-step guide to replicating this dashboard for any island cancer registry

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## 6.1 Step-by-Step Adaptation Checklist

Follow this sequence when setting up the dashboard for a new SIDS registry. Each step references the relevant section of this toolkit.

### Phase 1: Data Preparation

- Obtain incidence records in CSV format; verify all variables in Section 2.2
- Obtain mortality records; verify Section 2.3 variable list
- Download UN WPP population data for your country and year range (Section 2.4)
- Obtain administrative boundary shapefile from GADM (gadm.org) (Section 2.5)
- Extract or estimate sub-national population by 10-yr age band (Section 2.5)
- Run the data cleaning checklist in Section 2.6

### Phase 2: Configuration

- Clone or copy the app.R file into a new project folder
- Update incidence and mortality file paths and year ranges
- Replace shapefile path and NAME\_1 join column name
- Update standardize\_parish\_names() for your administrative unit names
- Set dashboard title, logo and branding colours (#253494 → your registry colour)
- Create hashed credentials using sodium::password\_store() (Section 4.4)

### Phase 3: Testing

- Run app locally: shiny::runApp("registry-dashboard/")
- Log in and verify all 8 modules load without errors
- Check incidence module: ASIR values should be >0 and site list populated
- Check mortality module: parish map should render with colour variation
- Check survival module: KM curve should display with reasonable shape
- Generate PowerPoint report: confirm all 15 slides render without NA errors
- Verify data quality indicators match your registry's known MV% and DCO%

### Phase 4: Deployment

- Add data/ to .gitignore before any version control operations
- Deploy to shinyapps.io or internal Shiny Server (Section 4.5)
- Set app visibility to Private; share login credentials only with authorised staff
- Schedule annual data refresh: update CSV files and re-deploy

## 6.2 Common Pitfalls & Solutions

**■ Parish names do not match shapefile**

**Solution:** Run `setdiff(unique(data$parish), shapefile$NAME_1)` after loading. All mismatches must be resolved in `standardize_parish_names()` or by renaming in one of the files. Common sources: "St." vs "Saint", accented characters, trailing spaces.

**■ KM survival curve is empty or flat**

**Solution:** Check that `time_months` (or equivalent) is  $> 0$  for all records. Ensure `dlc` (date of last contact) is correctly parsed with `dmy()`. Records with NA time will be silently dropped by `survfit()`.

**■ Projection model fails to converge**

**Solution:** This is expected for rare cancers. The framework automatically falls back to `lm()`. Check that at least 3 years of data exist for the selected site. Consider removing sites with fewer than 5 annual cases from the projection display.

**■ Map renders grey (no data)**

**Solution:** Indicates no matching join between shapefile `NAME_1` and `data$parish`. Re-run the `setdiff()` check above. Also verify that the selected cancer has cases in at least one parish.

**■ WPP population returns NA**

**Solution:** Sheet index mismatch: the `years` vector must match sheet order in `WPP.xlsx`. Verify with `readxl::excel_sheets("data/WPP.xlsx")` and reorder if needed.

**■ PowerPoint download produces blank slides**

**Solution:** Usually caused by a `ggplot` error inside `create_powerpoint_report()`. Run each plot-generating block interactively to isolate the failing chart. Wrap in `tryCatch()` to allow the remaining slides to render.



APPENDICES

Appendices

ICD-10 reference sites, glossary of terms and quick-reference card

Appendix A: Common IARC Cancer Site Labels (siteiarc)

| IARC Label (siteiarc)        | ICD-10 Code(s) | Sex    | Notes                                |
|------------------------------|----------------|--------|--------------------------------------|
| Lip, oral cavity & pharynx   | C00–C14        | Both   |                                      |
| Oesophagus                   | C15            | Both   |                                      |
| Stomach                      | C16            | Both   |                                      |
| Colorectum                   | C18–C21        | Both   | Colon + rectum combined              |
| Liver                        | C22            | Both   |                                      |
| Pancreas                     | C25            | Both   |                                      |
| Lung                         | C33–C34        | Both   |                                      |
| Melanoma of skin             | C43            | Both   |                                      |
| Breast (female)              | C50            | Female | Female only in IARC coding           |
| Cervix uteri                 | C53            | Female |                                      |
| Corpus uteri                 | C54–C55        | Female |                                      |
| Ovary                        | C56            | Female |                                      |
| Prostate                     | C61            | Male   |                                      |
| Testis                       | C62            | Male   |                                      |
| Kidney                       | C64–C66, C68   | Both   |                                      |
| Bladder                      | C67            | Both   |                                      |
| Brain & CNS                  | C70–C72        | Both   |                                      |
| Thyroid                      | C73            | Both   |                                      |
| Non-Hodgkin lymphoma         | C82–C86, C96   | Both   |                                      |
| Leukaemia                    | C91–C95        | Both   |                                      |
| Other and unspecified (O&U;) | C76–C80        | Both   | Excluded from most rate calculations |

Appendix B: Glossary

| Term                 | Definition                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------|
| ASIR                 | Age-Standardised Incidence Rate — incidence rate adjusted for age using WHO 2000 standard population, expressed per 100,000. |
| ASMR                 | Age-Standardised Mortality Rate — death rate adjusted for age using WHO 2000 standard, expressed per 100,000.                |
| Basis of Diagnosis   | Method by which a cancer was confirmed: Histology, Cytology, Lab, Clinical, DCO, etc.                                        |
| Cumulative Incidence | Probability of developing cancer between birth and age 74, assuming no competing deaths.                                     |
| DCO                  | Death Certificate Only — case registered solely from a death certificate; no prior cancer diagnosis in registry.             |

|                   |                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------|
| GLOBOCAN          | Global Cancer Observatory (IARC) — international cancer incidence and mortality estimates for 185 countries.         |
| IARC              | International Agency for Research on Cancer — WHO agency; sets coding standards for cancer registries.               |
| ICD-10            | International Classification of Diseases, 10th Revision — standard for coding disease diagnoses and causes of death. |
| Ill-defined Sites | Cancers coded to non-specific topography codes (C76, C80) indicating unknown or overlapping primary site.            |
| Kaplan-Meier      | Non-parametric method for estimating survival probability over time, accounting for censored observations.           |
| Log-rank Test     | Statistical test comparing survival distributions between two or more groups; produces a p-value.                    |
| MV%               | Microscopic Verification percentage — proportion of cases with histological or cytological confirmation.             |
| Negative Binomial | Count regression model that allows for overdispersion; used here for projecting cancer case counts.                  |
| Prevalence        | Number of living cancer survivors at a defined point in time (point prevalence).                                     |
| SIDS              | Small Island Developing States — UN-defined group of small island and low-lying coastal countries.                   |
| WPP               | UN World Population Prospects — official UN estimates of national population by age, sex and year.                   |
| WHO 2000 Standard | Reference population (100,000 persons) with fixed age distribution used for age-standardisation.                     |

## Appendix C: Quick-Reference Card

Key facts to keep at hand when running or presenting from the dashboard.

| Item                         | Value / Reference                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| WHO 2000 weights (18 groups) | 8860, 8690, 8590, 8470, 8220, 7930, 7610, 7150, 6590, 6040, 5380, 4550, 3720, 2960, 2210, 1520, 900, 600 (per 100,000) |
| Age group index              | 1 = 0–4 yrs ... 15 = 70–74 yrs ... 18 = 85+ yrs                                                                        |
| Cumulative incidence ages    | Age groups 1–15 only (0–4 to 70–74 years)                                                                              |
| Survival censoring window    | last_contact >= prevalence_date - 365 days                                                                             |
| NB model max iterations      | glm.control(maxit = 200)                                                                                               |
| Default projection horizon   | 5 years beyond last incidence year                                                                                     |
| Map CRS                      | EPSG:4326 (WGS84); transform from 3857 on load                                                                         |
| Map colour palette           | YlOrRd (ColorBrewer); NA parishes shown in grey (#808080)                                                              |
| MV% target (IARC)            | >75% for high-quality registry                                                                                         |
| DCO% target (IARC)           | <5% for high-quality registry                                                                                          |
| Ill-defined sites target     | <10%                                                                                                                   |
| Dashboard primary colour     | #253494 (BNR blue) — update for each registry                                                                          |
| Password hashing             | sodium::password_store() — bcrypt with random salt                                                                     |

### SIDS Cancer Registry Toolkit v1.0 · 2026

Based on the Barbados National Cancer Registry (BNR) R/Shiny Dashboard.

Designed for replication across Caribbean, Pacific, Indian Ocean, Atlantic and Mediterranean SIDS.

WHO 2000 Age-standardisation · IARC ICD-10 coding · UN World Population Prospects · R/Shiny open-source stack.